

Garmetix

Stage 8 Development Roadmap

From Stage 7M / Version 3.12.0 to production-ready v4.x

Current baseline	Stage 7M - Pre-v4.0 UI Naming and Menu Cleanup
Version / build	3.12.0 / GARMETIX-7M-20260611-3120
Branch / commit	Version3.0 / 6d440a5
Prepared	11 June 2026
Roadmap scope	Baseline stabilization, UI, access, returns, inventory, accounting, tests, integrations, and go-live

Purpose: Define the implementation sequence, boundaries, acceptance evidence, and production gates for the next development cycle while preserving the Stage 7M user experience and accounting architecture.

Source note: Stage 8A and Stage 8B incorporate the Stage 8 direction preserved locally from the web-chat planning history. The private 'Garmetix Web 2' conversation itself was not available as an exported or public transcript, so this report does not invent or quote unseen chat content. Stage 8C through Stage 8G consolidate the current repository audit and remaining engineering gaps.

Executive Summary

Garmetix already contains broad operational coverage across billing, purchase, inventory, accounting, banking, GST, HR, payroll, access control, backup, reporting, audit, and Oracle secondary synchronization. Stage 8 is therefore a hardening and completion program rather than a restart.

The implementation sequence begins with known baseline defects, then standardizes the user experience, strengthens access administration, formalizes purchase-return and inventory documents, improves accounting fidelity, introduces dependable automated testing, and finally validates external integrations and deployment.

Phase	Primary outcome	Business value	Release gate
P0	Stable baseline	Reliable dashboard, clean warnings, repeatable validation	No known response-body defect
8A	Consistent v4.0 UI	Faster daily work and fewer layout errors	All UI Audit pages reviewed
8B	Permission confidence	Correct role boundaries and administration	Role matrix passes
8C	Formal purchase returns	Traceable stock, GST, vendor and ledger settlement	Return lifecycle reconciles
8D	Authoritative inventory	Defensible stock and valuation	Movement ledger reconciles
8E	Accounting fidelity	Accurate split payments, banking and due balances	Journals and subledgers balance
8F	Automated assurance	Safer releases and fewer regressions	Critical suites pass
8G	Production go-live	Validated integrations, backup and deployment	Release checklist signed off

Delivery Principles

Preserve business boundaries: Regular accounting vouchers and Off Book Cash Vouchers remain separate.

Prefer formal documents: Transactions that affect stock, GST, vendor/customer balances, or audit must have traceable header/detail records.

Keep background flags internal: Party-ledger and bank-ledger linkage flags are derived by the system and are not user-entered controls.

Build for daily operation: Large master-detail workflows use full pages or wide workspaces, not narrow side drawers.

Prove completion: Every stage ends with build, runtime, reconciliation, permission, and visual evidence.

1. Baseline and Immediate Stabilization

Stage 7M contract: Preserve the grouped Nuxt UI shell, menu naming, current routes, permission-aware navigation, dark theme default, Off Book grouping, contextual Store/Company dashboards, and NUXT_PUBLIC_DASHBOARD_SHELL=legacy rollback.

Workstream	Implementation details	Completion evidence
Dashboard route	Correct the Minimal API mapping so GET /api/dashboard/home writes DashboardHomeDto to the response.	Authenticated request returns valid JSON and smart redirect works.
Compiler quality	Resolve nullable warnings in purchase receipt mapping and data-consistency number handling.	Release build completes without project warnings.
Build reliability	Eliminate external font metadata dependency or configure a reliable system-CA/local-font path.	Nuxt build succeeds without provider certificate failures.
Version identity	Keep API, frontend package, visible release, stage and build code synchronized.	System Info reports a full match.
Validation entry point	Create one current aggregate validation command for backend, frontend, static checks and smoke tests.	Single command produces a clear pass/fail report.
Docker runtime	Verify login, dashboard, workspace selection, role navigation, API health and database readiness.	Documented smoke run passes on the current Compose stack.

Exit gate

Stage 8 feature work begins only after the dashboard response defect is fixed, builds are repeatable, version identity matches, and the current Docker stack passes a basic authenticated smoke test.

8A

Full UI Audit and Standardization

Target v4.0 | Web Chat Stage 8 direction preserved locally

Objective: make every operational page feel like one coherent Nuxt UI application, with practical forms and predictable navigation for repeated store work.

Workstream	Implementation details	Completion evidence
Page framework	Standardize page title, context, primary action, filters, toolbar, table, pagination, loading, error and empty states.	Every route follows the approved page pattern.
Large forms	Move invoice, purchase, voucher, employee, payroll and other master-detail forms to dedicated pages or wide workspaces.	No critical workflow is constrained by a narrow drawer.
Responsive shell	Verify mobile drawer, collapsed icons, active groups, command palette, footer menus, workspace selector and topbar actions.	No overlap or inaccessible control at mobile, tablet or desktop sizes.
Tables and actions	Contain horizontal overflow, keep actions stable, wrap buttons cleanly and preserve readable columns.	Tables never break the shell or hide essential actions.
Notification UX	Add a useful action/notification surface for operational exceptions without exposing technical status noise.	Users can review and act on meaningful alerts.
Reusable components	Extract shared page-header, filter, form-section, master-detail and status patterns where duplication is material.	Reduced repeated markup without obscuring module logic.
Visible language	Remove stale stage/migration language and normalize capitalization, labels, help text and errors.	Business pages contain only user-relevant language.
Visual QA	Use /ui-audit plus Playwright screenshots at representative desktop and mobile viewports.	Every audit item is reviewed and recorded.

Module order

Sales and Billing; Purchase and Returns; Inventory and Product Master; Accounting and Banking; HR and Payroll; Reports and GST; CRM; Admin, Data, Maintenance and System.

8B

User, Role and Permission Hardening

Target v4.1 | Web Chat Stage 8 direction preserved locally

Objective: turn the existing access screens and policies into a verified administration system with clear scope, action rights and audit evidence.

Workstream	Implementation details	Completion evidence
User lifecycle	Audit create, edit, activate/deactivate, password reset and assigned workspace flows.	Admin can manage users without direct database work.
Role assignment	Validate Owner, Admin, PowerUser, Accountant, Remote Accountant, StoreManager, Salesman, HR and Payroll behavior.	Each role sees only approved menus and data.
Action rights	Enforce view, entry, edit and delete separately in both API and frontend.	StoreManager has view/entry but restricted edit/delete; delete remains Owner/Admin.
Admin isolation	Keep Company, Roles/Users, onboarding, seeding and administrative tools hidden and blocked for unauthorized roles.	Direct URL and API attempts are denied.
Scope rules	Validate company, store-group and store locking, defaults and workspace changes.	Transactions cannot escape assigned scope.
Audit trail	Record role, scope, status and password-administration changes.	Each administrative change has actor, time and target.
Automated matrix	Create integration and browser tests for menu, route, API and action permissions.	Permission suite passes for every supported role.

Important clarification: The user list and core access UI already exist. Stage 8B is an audit, refinement and test phase, not a claim that user administration must be created from nothing.

8C

Purchase Return and Vendor Settlement

Post-v4.1 business completion | Repository gap analysis

Objective: replace inferred purchase-return behavior with a formal, auditable document lifecycle tied to stock, GST, vendor balances and settlement.

Workstream	Implementation details	Completion evidence
Data model	Add PurchaseReturn and PurchaseReturnItem with original purchase references and immutable item/tax snapshots.	Return remains understandable even if masters change later.
Return workflow	Support partial quantities, reasons, validation against available return quantity, approval/cancellation and history.	No item can be over-returned.
Stock posting	Create linked outward stock movements and reverse them on controlled cancellation.	Return document and movement ledger reconcile exactly.
GST treatment	Calculate item-level HSN, CGST, SGST and IGST reversal for input tax credit.	GST return data matches return items.
Commercial note	Create/link vendor debit notes and provide A4/A5/PDF print actions.	Return, note and original purchase are mutually traceable.
Settlement	Support vendor refund receipt, balance adjustment, replacement and outstanding carry-forward.	Vendor subledger equals the chosen settlement path.
Accounting	Post balanced journals through the existing accounting service.	Journal, vendor ledger and bank/cash account reconcile.
Audit	Record create, approve, settle, cancel and reprint events.	Complete lifecycle visible from the return detail page.

8D

Inventory Documents and Valuation

Inventory authority program | Repository gap analysis

Objective: make inventory defensible through formal documents, an authoritative movement ledger, consistent valuation and reconciliation reports.

Workstream	Implementation details	Completion evidence
Adjustment document	Add header/items for damage, shortage, excess, write-off and correction with reason and approval.	Each adjustment has a source document.
Transfer document	Add source/destination stores, dispatch, receive, variance and in-transit states.	Both stores reconcile without duplicate stock.
Physical count	Add count sessions, snapshots, counted quantities, variances, approval and posting.	Posted variance links to movement and journal entries.
Movement authority	Derive stock balance from StockMovement rather than cumulative purchase/sold counters.	Movement sum equals displayed and report stock.
Valuation	Implement weighted-average cost first and document treatment of returns, transfers and adjustments.	Valuation report reproduces ledger value.
Accounting	Post shortage, excess, write-off and transfer implications where required.	Inventory value and accounting remain aligned.
Analytics	Add ageing, low-stock risk, slow-moving, valuation and reconciliation reports.	Reports support store and company scopes.
Concurrency	Test sequence safety, stock locks, negative-stock policy and competing transactions.	Concurrent operations cannot oversell or corrupt stock.

8E

Accounting, Banking and Payment Hardening

Financial fidelity program | Repository gap analysis

Objective: ensure every payment instrument, allocation and derived ledger relationship is complete, balanced and auditable under Indian accounting practice.

Workstream	Implementation details	Completion evidence
Split payment posting	Post one accounting line per cash, card, UPI, bank, cheque, advance, credit note, loyalty or store-credit row.	Invoice payment rows reconcile to journal lines.
Instrument details	Persist bank account, card/UPI reference, cheque number/date, issuer, depositor, beneficiary and purpose.	Audit can identify who paid whom, how and why.
Value application	Prevent duplicate use of advances, credit notes, loyalty and store credit.	Each source has an application ledger and remaining balance.
Bank reconciliation	Import/enter statement rows and match, partially match, unmatched or adjust transactions.	Book balance and statement balance reconcile.
Cheque lifecycle	Track issued, received, deposited, cleared, bounced, cancelled and replaced states.	Cheque log and bank ledger remain synchronized.
Year controls	Add financial-year lock, reopening authority and posting-date validation.	Locked periods reject unauthorized changes.
Journal control	Strengthen debit/credit balancing, reversal and duplicate-reference rules.	No unbalanced journal can be posted.
Derived ledgers	Keep Party and Bank Account ledger creation/linking internal and idempotent.	Users never edit internal flags; links survive rename/edit.
Management views	Add customer/vendor dues, cash/payment summary and store-group comparisons.	Company dashboard exposes actionable financial positions.

8F

Deep Audit and Automated Testing

Release assurance | Repository gap analysis

Workstream	Implementation details	Completion evidence
Change history	Store before/after JSON for material entities with actor, reason, time and correlation reference.	Auditors can reconstruct important changes.
Backend tests	Add xUnit unit and API integration tests for domain rules and endpoints.	Critical modules have repeatable automated coverage.
Database tests	Exercise PostgreSQL transactions, locks, sequences, migrations and rollback behavior.	Concurrency and migration scenarios pass.
Financial tests	Cover journal balance, GST, returns, payroll, advances, dues and reconciliation.	Expected ledgers match golden test cases.
Frontend tests	Add Playwright flows for login, billing, purchase, returns, stock, payroll and administration.	Critical user journeys pass in CI/local validation.
Recovery tests	Restore a backup into a disposable database and verify manifests, checksums and key counts.	Recovery drill has recorded evidence.

8G

External Integration and Production Go-Live

Production release | Repository gap analysis

Workstream	Implementation details	Completion evidence
Email	Validate SMTP reset and payslip delivery with production sender/domain.	Delivered messages and failures are logged.
Google Drive	Validate service account, folder permissions, upload, download and restore.	Cloud backup and recovery drill passes.
GST provider	Configure licensed GSTIN lookup credentials and validate production response mapping.	Verified party data and mismatch alerts work.
Oracle Cloud	Validate wallet/TNS, push/pull, review/apply, dead letters and one real external app.	Ownership and final auto-apply policy are signed off.
Clean installation	Run fresh Docker installation, migrations, onboarding, seed and first login.	New installation reaches healthy operational state.
Production security	Configure HTTPS, secrets, CORS, retention, scheduled backups and log rotation.	Preflight reports no critical production issue.
Acceptance	Run role matrix, module smoke tests, reports, printing, backup/restore and accounting reconciliation.	Production release checklist is approved.

2. Delivery Sequence and Governance

Order	Package	Depends on	Primary verification	Decision gate
1	P0	Stage 7M baseline	Build + authenticated API smoke	Baseline accepted
2	8A	P0	Visual audit + responsive screenshots	v4.0 UI accepted
3	8B	8A patterns	Role and permission matrix	v4.1 access accepted
4	8C	P0 accounting/stock	Return, GST and vendor reconciliation	Return lifecycle accepted
5	8D	8C movement rules	Stock and valuation reconciliation	Inventory authority accepted
6	8E	8C/8D postings	Journal, payment and banking reconciliation	Financial controls accepted
7	8F	Stable business rules	Automated suites + restore drill	Release candidate accepted
8	8G	All prior stages	Real integrations + go-live checklist	Production approved

Stage note discipline

Each implementation package will add its plan to docs/planning, its durable technical notes to docs/modules, and its release notes plus validation evidence to the matching docs/stages folder. Issues discovered during development will be recorded in docs/planning/ISSUES.md before being marked fixed.

Definition of done

Workstream	Implementation details	Completion evidence
Code	Backend, frontend, migrations and configuration are complete and scoped to the stage.	Peer-readable diff and no unrelated refactor.
Build	Release backend and production Nuxt builds complete.	Build log recorded.
Runtime	Docker services are healthy and the changed workflow is exercised.	Smoke evidence recorded.
Data	Migrations and older-volume repair paths are validated where applicable.	Fresh and upgrade paths pass.
Security	Role, scope and action permissions are checked.	Unauthorized API and route attempts fail.
Business	Stock, GST, payroll or accounting effects reconcile as applicable.	Expected totals and ledger links match.
UX	Desktop and mobile layouts are visually inspected.	No overlap, clipping or unusable form.
Documentation	Roadmap, TODO, issue register, stage notes and validation log are updated.	Canonical docs remain current.

3. Principal Risks and Controls

Risk	Impact	Control
UI migration causes workflow regression	Daily billing or purchase work slows or breaks	Module-by-module rollout, existing routes preserved, legacy shell retained until acceptance
Return documents diverge from stock/GST/accounting	Incorrect vendor balance or tax reporting	Single transactional posting service plus reconciliation tests
Movement-ledger migration changes displayed stock	Unexpected stock variance	Pre-migration comparison report and controlled repair path
Permission mismatch between client and API	Unauthorized access or false denial	Server-first policies plus automated role matrix
External integration credentials are unavailable	Stage cannot be production-validated	Keep provider adapters configurable and mark credential-bound gates explicitly
Automated tests arrive too late	Hardening changes create regressions	Add tests alongside each Stage 8C-E business rule, then consolidate in 8F

Source and provenance

This roadmap was compiled from the Stage 7M repository baseline, docs/planning/CURRENT-ROADMAP.md, docs/planning/MASTER-TODO.md, docs/planning/ISSUES.md, docs/planning/TODO-RECONCILIATION.md, Stage 7 implementation notes, local build/runtime audit results, and the Stage 8A/8B direction retained in the locally preserved web-chat planning history.

The private ChatGPT conversation named 'Garmetix Web 2' was not available in the local workspace as an export and was not accessible through a public shared link. Consequently, this document includes all available locally preserved Stage 8 content but does not claim verbatim or complete access to that private conversation.

Recommended first implementation package: Priority 0 Baseline Stabilization: fix the empty dashboard response, remove compiler warnings, stabilize Nuxt font/build behavior, add aggregate validation, and run the authenticated Docker smoke test. This establishes a trustworthy base before v4.0 UI work.